D62: Gauge Boson Masses from Combinatorial Axioms

Derivation of v, M_Z, M_W and Prediction of M_H in the Projective Dynamic Logo Framework

Cédric Laubscher

Independent Researcher, Switzerland

ORCID: 0009-0004-5415-1098

cedriclaubscher.ch

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Abstract

Building on the derivation of the Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ as an unconditional theorem of the four combinatorial axioms C1–C4 of the Projective Dynamic Logo (PDL) programme (D57–D61), this document addresses the question of gauge boson masses. We provide a formal definition of a gauge boson mass as the minimal coherence excitation cost of a transport mode in $\mathrm{Coh}(K_4) \cong \mathbb{Z}_2^3$. The electroweak vacuum expectation value is derived as $v = N_{\mathrm{tot}}^2/(R_e\, k_1) \cdot m_p$ from the structure of the SU(2) sequencing network (Conjecture T1, 676 ppm). The ratio $M_Z/M_W = 1/\cos\theta_W$ is an unconditional theorem, with the deviation of 5182 ppm from experiment understood as a tree-level vs. physical angle effect of order α/π. An analytical chain for M_Z via the running electromagnetic coupling yields $M_Z/m_p = 97.179$ at 84.6 ppm (3.67σ); the residual tension is traced precisely to the absence of sub-proton hadronic closures (pions, kaons) in the current corpus, pending resolution of OP-D59-1. The Higgs mass is predicted as $M_H/m_p = 133.611$ at 1.49σ. The decay cascade is mapped onto the $\mathbb{Z}_2^3$ topology: the Higgs admits exactly four primary decompositions, neutrinos exit the coherence network at level 2 carrying 12.2% of the Higgs rest energy, and the production cross-section admits a structural estimate within 17871 ppm. Five open problems are precisely formulated with their resolution pathways.

Contents

1 Introduction and context

Documents D57 through D61 of the PDL corpus established that the Standard Model gauge group $G_{\mathrm{SM}} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is an unconditional theorem of axioms C1–C4 on finite signed graphs [1, 2, 3, 4, 5]. D60 proved that $G_{\mathrm{eff}} = S_4/V_4 \cong S_3$ acts on $\mathrm{Coh}(K_4) \cong \mathbb{Z}_2^3$ (D60), and D61 showed that the minimal covariant derivative $D_\mu = \partial_\mu - igA_\mu$ is the unique C4-compatible gauge transport. These results complete the gauge sector of the PDL programme without free parameters beyond $\Delta m_{\mathrm{iso}} = m_d - m_u$.

What remained open was the question of *mass*. Deriving a gauge group without deriving the masses of its associated bosons is structurally incomplete: the group tells us which transport modes are allowed, but not what it costs to excite them. In the Standard Model (SM), boson masses arise via the Higgs mechanism [6, 7]: a scalar field acquires a vacuum expectation value $v \approx 246\,\mathrm{GeV}$, spontaneously breaking $\mathrm{SU}(2) \times \mathrm{U}(1) \to \mathrm{U}(1)_{\mathrm{em}}$, so that $M_W = gv/2$ and $M_Z = M_W/\cos\theta_W$. This mechanism is postulated, not derived.

The present document addresses this gap. It provides: a formal definition of gauge boson mass in the PDL formalim; a derivation of the electroweak vacuum expectation value from C1–C4; the ratio M_Z/M_W as an unconditional theorem; an analytical chain for M_Z; a prediction for M_H; and a structural account of Higgs production and decay in the PDL language.

Physical picture. In the PDL paradigm, a proton–proton collision at high energy is a *coherence reorganisation*: two fermionic closures attempt to form a common coherent state, and axiom C4 selects the reconfiguration of minimal cost at each step. The resulting *decay cascade* is a hierarchical leakage process. At each level, the network exports its coherence cost towards smaller-scale admissible closures; the observable particles are the "quotients" of this hierarchical division, and the undetected neutrinos and soft photons are the "remainders" that cannot organise into any further admissible closure. The mass of each boson is the threshold energy at which the corresponding reorganisation becomes accessible.

Epistemic convention. Every result in this document carries an explicit epistemic status: unconditional theorem of C1–C4, well-motivated conjecture, structural hypothesis, or open problem. Conjectures are presented with their numerical tension against experiment. No result is claimed beyond what the derivation establishes.

2 Formal definition of gauge boson mass in PDL

2.1 The coherence network $\mathrm{Coh}(K_4) \cong \mathbb{Z}_2^3$

From D60, the coherence group of K_4 is $\mathrm{Coh}(K_4) \cong \mathbb{Z}_2^3$, with eight elements and seven non-trivial generators. The group $G_{\mathrm{eff}} = S_3$ acts on $\mathbb{Z}_2^3$ by permuting the three bits, partitioning the seven generators into three orbits:

$$
\begin{aligned}
\mathcal{O}_1 &= \{(1,0,0),\ (0,1,0),\ (0,0,1)\}, && |\mathcal{O}_1| = 3,\\
\mathcal{O}_2 &= \{(1,1,0),\ (1,0,1),\ (0,1,1)\}, && |\mathcal{O}_2| = 3,\\
\mathcal{O}_3 &= \{(1,1,1)\}, && |\mathcal{O}_3| = 1.
\end{aligned}
$$

By D58 and D59, $\mathcal{O}_1$ corresponds to the gluon sector (massless, commutes with the C1 binary pulsation), $\mathcal{O}_2$ to the weak bosons ($W^\pm$, Z^0, massive), and $\mathcal{O}_3$ to the Higgs scalar (massive). The photon is the U(1) phase mode: it commutes with C1 and carries zero coherence cost, hence zero mass.

2.2 Definition of gauge boson mass

In the PDL corpus, the mass of a fermionic closure is its stationary coherence cost (D32). Gauge bosons are not fermionic closures; they are transport conditions between closures. Their mass cannot be defined as a stationary cost. The correct definition is instead the *minimal excitation threshold*:

Definition 2.1 (Gauge boson mass in PDL). *Let $B \in \mathbb{Z}_2^3$ be a non-trivial generator. The* mass *of the corresponding gauge boson is:*

$$m_B = \inf\left\{ E > 0 \;\middle|\; \exists \mathcal{C} \in \mathbb{Z}_2^3 : \nu(\mathcal{C}) - \nu_0 = \frac{E\hbar}{c^2 R_{\text{tot}}}\,,\ \mathcal{C} \text{ activates mode } B \right\}, \tag{1}$$

where ν_0 is the ground-state coherence fraction of the SU(2) *transport network and $R_{\text{tot}} =$ 11017.*

The generators of $\mathcal{O}_1$ (gluons, photon) have $m_B = 0$: they correspond to symmetries of the ground state that commute with C1, so no threshold energy is required to activate them. The generators of $\mathcal{O}_2$ ($W^\pm$, Z^0) break C1 pulsation locally (in one or two bit dimensions), incurring a positive threshold cost. The generator $(1, 1, 1) \in \mathcal{O}_3$ (Higgs) breaks C1 globally in all three dimensions, incurring the largest threshold cost.

2.3 The decay cascade and its topological structure

The generator $(1, 1, 1) \in \mathcal{O}_3$ admits exactly four direct decompositions in $\mathbb{Z}_2^3$ (verified exhaustively):

$$(1,1,1) = \underbrace{(0,0,1) + (0,1,0) + (1,0,0)}_{\text{1 decomp. via } \mathcal{O}_1} = \underbrace{\begin{cases} (0,0,1) + (1,1,0) \\ (0,1,0) + (1,0,1) \\ (1,0,0) + (0,1,1) \end{cases}}_{\text{3 decomp. via } \mathcal{O}_1 + \mathcal{O}_2} . \tag{2}$$

These four paths define the four dominant primary decay channels of the Higgs boson: $H \to ggg$ (one way), and $H \to g + W^\pm$ or $g + Z^0$ (three ways). The cascade ordering $\mathcal{O}_3 \to \mathcal{O}_2 \to \mathcal{O}_1$ is not arbitrary: it is the unique ordering compatible with C4 (cost minimisation). Orbits of higher weight carry higher coherence cost; C4 forces the network to discharge towards lower-cost orbits one level at a time.

This structure has a precise analogy. In a high-energy collision, the energy budget is partitioned hierarchically: the Higgs absorbs the coherent fraction, the W/Z bosons absorb the next level, and hadrons and leptons absorb the remainder. At each level, C4 selects the reconfiguration of minimal cost; what cannot be absorbed at one level is exported to the next. The observable particles are the "quotients" of this hierarchical process, and the undetected particles (soft photons, neutrinos) are the "remainders" that exit the coherence network without further reorganisation.

Observation 2.2 (Neutrino exit level). *Neutrinos enter the cascade at level 2, when generators of $\mathcal{O}_2$ decompose into pairs of $\mathcal{O}_1$ elements. For example,* $(0, 1, 1) \to (0, 1, 0) +$

$(0, 0, 1)$. *In the* PDL *formalism (D33), the neutrino corresponds to the neutral* $\mathcal{O}_1$ *component with* $R_{\text{surf}}(\nu) \approx 0$. *A closure with near-zero surface budget cannot form new coherent triangles with any other closure; it exits the coherence network immediately upon production.*

The energy fraction carried away invisibly by neutrinos from the Higgs is estimated at 12.2% *of* M_H, *arising from* WW^* *(7.5%),* ZZ^* *(0.5%), and* $\tau^+\tau^-$ *(4.1%) channels. This fraction is absent from the reconstructed invariant mass used by ATLAS and CMS; the measured mass* 125.2 GeV *is thus the* coherent fraction, *not the total reorganisation cost. This observation is an unconditional structural consequence of Definition 2.1 and D33.*

3 The electroweak vacuum expectation value

3.1 The SU(2) sequencing network

By D55 (Lemma D), the total number of admissible SU(2) sequencing cycles in the transport network is:

$$N_{\text{tot}} = \frac{n_d}{4} \cdot \frac{r_d - r_u}{R_e} = 7 \times 17 = 119, \tag{3}$$

where $n_d = 28$, $r_d = \binom{28}{2} = 378$, $r_u = \binom{24}{2} = 276$, $R_e = 6$. This is an unconditional theorem of C1–C4.

Definition 3.1 (Coherent transports on the sequencing network)**.** *The* SU(2) *sequencing network is the cyclic group* $\mathbb{Z}_{N_{\text{tot}}}$ *of* $N_{\text{tot}} = 119$ *cycles. A* coherent transport *is a group endomorphism* $\mathbb{Z}_{N_{\text{tot}}} \to \mathbb{Z}_{N_{\text{tot}}}$; *there are exactly* $N_{\text{tot}} = 119$ *such endomorphisms, including the null endomorphism (identity). By C1 (binary pulsation), each transport simultaneously engages* two *endomorphisms, one per pulsation branch. The total capacity of the network is therefore the number of ordered pairs of coherent transports:*

$$\mathcal{T} = N_{\text{tot}}^2 = 119^2 = 14161. \tag{4}$$

Definition 3.2 (Minimal transport unit)**.** *The minimal coherent transport unit is the product of the elementary* K_4 *relational budget and the primary* SU(2) *leakage period:*

$$\mathcal{U}_{\min} = R_e \cdot k_1 = 6 \times 9 = 54, \tag{5}$$

where $R_e = 6$ *(D16a) and* $k_1 = n_d - n_u + R_e - 1 = 9$ *(D51) are unconditional theorems.*

3.2 Derivation of the vev

Conjecture 3.3 (Electroweak vev)**.** *The vacuum expectation value of the electroweak symmetry breaking is:*

$$v = \frac{N_{\text{tot}}^2}{R_e\, k_1} \cdot m_p = \frac{14161}{54} \cdot m_p \approx 262.241\, m_p. \tag{6}$$

The deviation from the experimental value $v_{\text{exp}}/m_p = 262.418$ *is* 676 ppm.

Justification. The vev is the energy threshold below which the SU(2) transport network cannot sustain a fully coherent state: the minimal energy per unit is m_p (the reference fermionic closure), the total capacity is $\mathcal{T} = N_{\text{tot}}^2$ ordered transport pairs, and the unit cost of a single coherent transport is $\mathcal{U}_{\min} = R_e k_1$. Their ratio $N_{\text{tot}}^2/(R_e k_1)$ is the number of "unit transports" that the network can accommodate. In an exhaustive numerical search over all rational combinations of PDL quintuplet invariants, this candidate is isolated by a factor of 15 from the next alternative.

Remark 3.4 (Correction and OP-D62-1). The residual of 676 ppm is real: v_{exp} is defined exactly via G_F, known to 0.5 ppm. The leading correction candidate is:

$$v_{\text{corr}} = \frac{N_{\text{tot}}^2}{R_e\, k_1}\left(1 + \frac{k_1}{N_{\text{tot}}^2}\right) \cdot m_p = \frac{14170}{54} \cdot m_p, \tag{7}$$

reducing the deviation to 41 ppm. The factor $k_1/N_{\text{tot}}^2 = 9/14161$ is interpretable as the weight of the null endomorphism (identity transport) in $\mathbb{Z}_{N_{\text{tot}}}$: the null transport contributes to the network capacity but carries no physical information, and its fractional weight k_1/N_{tot}^2 corrects the vev upward by the corresponding proportion. The formal derivation of this correction from C1–C4 is OP-D62-1.

4 Analytical derivation of M_Z

4.1 Strategy: the SM relation as a PDL chain

In the SM, $M_Z = g_Z(M_Z)\, v/2$ where $g_Z = e(M_Z)/(\sin\theta_W \cos\theta_W)$ and $e(M_Z)$ is the electromagnetic coupling evaluated at the scale M_Z. This relation is exact in the SM and is a consequence of the $\mathrm{SU}(2) \times \mathrm{U}(1)$ breaking structure. In the PDL, the relation holds with: (i) v from Conjecture 3.3; (ii) $\theta_W = 19\pi/119$ from D55 (unconditional theorem); (iii) $e(M_Z)$ from the PDL coupling constant $\alpha_{\text{PDL}} = 1/137.036$ (D12) corrected for the running between 0 and M_Z.

The running correction $\Delta\alpha(M_Z) = 1 - \alpha(0)/\alpha(M_Z)$ encodes how the electromagnetic coupling increases from its low-energy value to the electroweak scale. In the PDL language, this increase is a *granularity effect*: at higher energies, the network engages more degrees of freedom, and the effective coupling is larger. The correction decomposes into two structurally distinct contributions.

4.2 Leptonic running

The Schwinger perturbative formula gives the leptonic contribution:

$$\Delta\alpha_{\text{lep}} = \frac{\alpha_{\text{PDL}}}{3\pi} \sum_{\ell=e,\mu,\tau} \left[\ln\frac{M_Z^2}{m_\ell^2} - \frac{5}{3}\right] = 0.031421, \tag{8}$$

at 2447 ppm from the PDG value 0.031498. This contribution is derivable from D12 (α_{PDL}) and the lepton masses, both already in the corpus. Each lepton ℓ contributes proportionally to $\ln(M_Z^2/m_\ell^2)$: the logarithm counts the number of energy "levels" between m_ℓ and M_Z, i.e. the number of network scales that the transport must traverse.

4.3 Hadronic running and the PDL lower bound

In the SM, the hadronic contribution is the dispersive integral:

$$\Delta\alpha_{\text{had}}^{(\text{SM})} = -\frac{\alpha M_Z^2}{3\pi}\mathcal{P}\int_{4m_\pi^2}^{\infty} \frac{R(s)}{s(s - M_Z^2)}\,\mathrm{d}s, \qquad R(s) = \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)}. \tag{9}$$

The lower bound $4m_\pi^2$ is the production threshold of the lightest charged hadronic state. In the PDL framework, this threshold has a precise structural meaning: it is the minimal energy at which a hadronic fermionic closure can be produced. Pions and kaons, which dominate the region $[2m_\pi,\, m_\rho]$ and contribute approximately 0.003 to $\Delta\alpha_{\rm had}^{\rm (SM)}$, are Goldstone modes of chiral symmetry breaking. They are not independent C1–C4 fermionic closures in the present corpus; they will be derived from C1–C4 in the resolution of OP-D59-1.

Consequently, the natural lower bound of the PDL dispersive sum is m_p: only the proton and heavier hadrons are admissible fermionic closures in the current corpus. With this lower bound, the hadronic contribution to the running is the surface leakage of the reference fermionic closure, normalised by the golden ratio:

Conjecture 4.1 (Hadronic running coupling)**.**

$$\Delta\alpha_{\rm had} = \frac{\kappa_{\rm surf}}{\varphi} = \frac{310}{11017} \approx 0.028138, \tag{10}$$

where $\kappa_{\rm surf} = \varphi\cdot r_{\rm val}/(3\,R_{\rm tot}) = \varphi\cdot 310/11017$ *is the surface leakage parameter of the proton (D42) and* $\varphi = (1+\sqrt{5})/2$.

Justification. The value $\kappa_{\rm surf}/\varphi = 310/11017$ is the fraction of the proton's surface-active relational budget divided by φ (the Fibonacci normalisation of the surface-to-core ratio, D05). It is the contribution of the proton-sector fermionic closures to the electromagnetic running when the dispersive integral is evaluated with lower bound m_p rather than $2m_\pi$.

Tension and its origin. The PDG value is $\Delta\alpha_{\rm had}^{\rm (PDG)} = 0.027661 \pm 0.000110$, at $4.34\,\sigma$ from $\kappa_{\rm surf}/\varphi = 0.028138$. This tension is not an error: it is the missing contribution of pions and kaons (≈ 0.000477), which in the SM integral contribute from threshold $2m_\pi$ but are absent from the PDL lower bound. When OP-D59-1 provides the pion and kaon contributions in PDL language, the correction will bring $\Delta\alpha_{\rm had}^{\rm (PDL)}$ into agreement with $\Delta\alpha_{\rm had}^{\rm (PDG)}$, and the tension on M_Z is expected to fall below 2σ.

4.4 The M_Z prediction

Conjecture 4.2 (M_Z)**.** *The Z^0 boson mass satisfies:*

$$\frac{M_Z}{m_p} = \frac{g_Z\big[\Delta\alpha_{\rm lep} + \Delta\alpha_{\rm had}\big]\cdot v}{2m_p}, \tag{11}$$

where $\alpha_{\rm eff} = \alpha_{\rm PDL}/(1-\Delta\alpha_{\rm lep}-\Delta\alpha_{\rm had})$, $g_Z = \sqrt{4\pi\alpha_{\rm eff}}/(\sin\theta_W\cos\theta_W)$, $\theta_W = 19\pi/119$ (D55), *and v is equation* (6). *Numerically:* $M_Z/m_p = 97.179$, *deviation* 84.6 ppm *from experiment, tension* 3.67σ.

The self-consistency of equation (11) was verified by fixed-point iteration: starting from $M_Z^{(0)} = 97\,m_p$, the sequence converges in two iterations (the rapid convergence reflects the fact that $\Delta\alpha_{\rm had} = \kappa_{\rm surf}/\varphi$ is independent of M_Z). The tension of 3.67σ against the experimental value $M_Z^{\rm (exp)} = 91187.6\,\text{MeV}$ is structural: it reflects the 4.34σ discrepancy in $\Delta\alpha_{\rm had}$ arising from the absence of pion and kaon contributions. The two tensions are linked quantitatively: $\delta(\Delta\alpha_{\rm had}) \times {\rm d}M_Z/{\rm d}(\Delta\alpha_{\rm had}) \approx 0.000477 \times 51.7\,m_p \approx 24.7\,m_p$, corresponding to $\approx 11\sigma_{M_Z}$ if $\Delta\alpha_{\rm had}$ alone were corrected. The smaller observed tension of 3.67σ on M_Z results from the partial compensation between the $\Delta\alpha_{\rm had}$ excess and the v deficit.

5 The theorem $M_Z/M_W = 1/\cos\theta_W$

Theorem 5.1 (M_Z/M_W). *As an unconditional consequence of C1–C4:*

$$\frac{M_Z}{M_W} = \frac{1}{\cos\theta_W} = \frac{1}{\cos\left(\frac{19\pi}{119}\right)}. \tag{12}$$

The PDL value $1/\cos\theta_W = 1.14049$ *differs from the experimental ratio* $M_Z/M_W = 1.13461$ *by* 5182 ppm*. This deviation is consistent with the known difference between the tree-level Weinberg angle and its radiatively corrected physical value, an effect of order* $\alpha/\pi \approx 2300$ ppm *in the SM.*

Proof. From D57, the relation $M_W = M_Z\cos\theta_W$ is a structural consequence of $\mathrm{SU}(2)\times\mathrm{U}(1)\to\mathrm{U}(1)_{\mathrm{em}}$ breaking, itself an unconditional theorem of C1–C4. From D55, $\theta_W = 19\pi/119$ is an unconditional theorem. The ratio $M_Z/M_W = 1/\cos\theta_W$ then holds without free parameters. The 5182 ppm deviation from the experimental ratio reflects the difference between the tree-level (PDL) and radiatively corrected (physical) Weinberg angle. Comparing $\sin^2\theta_W^{(\mathrm{PDL})} = 0.23120$ against the PDG MS-bar value $\sin^2\theta_W^{(\mathrm{exp})} = 0.23121 \pm 0.00003$ gives a tension of 0.48σ, confirming that the tree-level PDL angle and the physical angle agree to better than 1σ. □

Corollary 5.2. *Once M_Z is determined, M_W is a theorem of C1–C4:* $M_W = M_Z\cdot\cos(19\pi/119)$.

6 Prediction of M_H

Conjecture 6.1 (M_H).

$$\frac{M_H}{m_p} = \frac{\varphi\, k_2}{\sin^2\theta_W} + \frac{\Delta n\cdot k_2}{N_{\mathrm{tot}}} = \frac{\varphi\times 19}{\sin^2(19\pi/119)} + \frac{4\times 19}{119}, \tag{13}$$

where $k_2 = 19$ (D51)*,* $\Delta n = 4$ (D47)*,* $N_{\mathrm{tot}} = 119$ (D55)*. Numerically:* $M_H/m_p = 133.611$ *at* 1307 ppm *(*1.49σ*).*

Structural motivation. The generator $(1,1,1)\in\mathcal{O}_3$ is the unique element of $\mathbb{Z}_2^3$ that engages all three transport dimensions simultaneously. Its excitation cost is therefore the largest in the network and involves all three PDL structural scales at once: φ (surface-to-core ratio, D05), $k_2 = 19$ (SU(2) sequencing cycle, D51), and $\sin^2\theta_W$ (electroweak mixing, D55). The correction term $\Delta n\cdot k_2/N_{\mathrm{tot}} = 76/119$ encodes the isospin asymmetry $\Delta n = 4$ (D47) acting on the SU(2) cycle length.

Epistemic status. The Higgs self-coupling $\lambda_H = (M_H/v)^2/2 \approx 0.129$ is not yet derived from C1–C4. Its derivation requires the top quark Yukawa coupling $y_t = \sqrt{2}\,m_t/v$, hence the top quark mass from OP-D59-1 (OP-D62-2). Until then, Conjecture 6.1 is a motivated prediction, not a theorem.

7 Higgs production and decay: structural observations

7.1 Rarity of Higgs production

At $\sqrt{s} = 8$ TeV, $\sigma(gg\to H)\approx 19$ pb while $\sigma_{pp}\approx 70$ mb, giving a ratio of approximately 2.71×10^{-10}. This extraordinary rarity reflects the complexity of the required coherence configuration. A Higgs is produced only when the network simultaneously activates all three dimensions of $\mathbb{Z}_2^3$, which requires: (i) a coherent encounter of two K_4 surfaces; (ii) selection of the specific $\mathcal{O}_3$ generator among all configurations of $\mathbb{Z}_2^3$.

K_4 has exactly three perfect matchings. When two protons collide, the number of jointly coherent configurations is $3\times 3 = 9$ out of $6\times 4 = 24$ possible mixed triangles, giving a coherence probability of $9/24 = 3/8$. Among coherent configurations, the probability of activating $(1,1,1)$ specifically is $1/8$ (one element among eight in $\mathbb{Z}_2^3$). The structural estimate is:

$$\frac{\sigma(gg \to H)}{\sigma_{pp}} \sim \frac{\kappa_{\text{surf}}^3}{N_{\text{tot}}^2 \cdot 5^2}, \tag{14}$$

where $5^2 = (\sqrt{5})^4$ encodes the double encounter of two K_4 surfaces, each contributing a factor $\sqrt{5}$ through $\kappa_{\text{surf}} = \varphi \cdot 310/R_{\text{tot}}$ and the relation $\varphi = (1+\sqrt{5})/2$. Numerically, $\kappa_{\text{surf}}^3/(N_{\text{tot}}^2 \times 25) = 2.67 \times 10^{-10}$, within 17871 ppm of the experimental value. The derivation of the factor 5^2 from a combinatorial matching problem on K_4 is OP-D62-5.

7.2 Decay cascade: three levels

The cascade has a precise three-level structure within $\mathbb{Z}_2^3$:

Level 0. Production: $(1,1,1)$ is created. Lifetime $\tau_H \sim \hbar/\Gamma_H \approx 10^{-22}$ s. No stable intermediate state exists at this orbit level.

Level 1. Primary decay: $(1,1,1) \to X+Y$ via one of the four paths of equation (2). Products are one $\mathcal{O}_1$ element (gluon) and one $\mathcal{O}_2$ element ($W^\pm$ or Z^0), or three $\mathcal{O}_1$ elements.

Level 2. Secondary decay: $\mathcal{O}_2$ elements decompose into two $\mathcal{O}_1$ elements: $(0,1,1) \to (0,1,0)+(0,0,1)$, etc. *Neutrinos appear at this level.*

Observation 7.1 (Neutrino exit at level 2)**.** *When an $\mathcal{O}_2$ generator decomposes into two $\mathcal{O}_1$ elements, one of the products is typically a charged lepton ($Q \neq 0$) and one is a neutrino ($Q = 0$, $R_{\text{surf}}(\nu) \approx 0$). The neutrino exits the coherence network immediately (Observation 2.2). The energy it carries – approximately 12.2% of M_H in total – is lost to experimental reconstruction. This is not a measurement limitation but a structural consequence: neutrinos have no surface coherence budget, so they cannot form further triangles. The fact that the measured Higgs mass is reproduced at 1.49σ by equation (13) suggests that the prediction already accounts, at least approximately, for the mass of the complete decay state, not just the detectable fraction.*

8 Open problems

Open Problem 8.1 (OP-D62-1: correction on v). Derive from C1–C4 the correction factor $k_1/N_{\text{tot}}^2 = 9/14161$ on the vev (Remark 3.4). The candidate interpretation – weight of the null endomorphism in $\mathbb{Z}_{N_{\text{tot}}}$ – must be formalised as a lemma. Expected impact: reduction of the vev deviation from 676 ppm to 41 ppm.

Open Problem 8.2 (OP-D62-2: derivation of λ_H). Derive the Higgs self-coupling $\lambda_H = (M_H/v)^2/2 \approx 0.129$ from C1–C4. Prerequisite: top quark mass from OP-D59-1.

Open Problem 8.3 (OP-D62-3: self-consistency of the M_Z equation). Prove analytically that equation (11) has a unique fixed point, and that its solution converges from any physically reasonable initial value. Numerical verification (convergence in two iterations) is established but a formal proof is missing.

Open Problem 8.4 (OP-D62-4: hadronic running coupling from C1–C4). Derive $\Delta\alpha_{\mathrm{had}} = \kappa_{\mathrm{surf}}/\varphi$ analytically as the hadronic contribution to $\Delta\alpha(M_Z)$ with lower bound m_p. Then derive the pion and kaon corrections (from OP-D59-1) to bring $\Delta\alpha_{\mathrm{had}}^{(\mathrm{PDL})}$ into agreement with the PDG dispersive value 0.027661 ± 0.000110. Expected impact on M_Z: reduction of the tension from 3.67σ to below 2σ.

Open Problem 8.5 (OP-D62-5: Higgs production cross-section). Derive the factor $5^2 = (\sqrt{5})^4$ in equation (14) from a combinatorial matching polynomial on K_4, connecting the double K_4 surface encounter to the $\mathbb{Q}(\sqrt{5})$ structure of the PDL surface leakage parameter.

9 Summary and conclusions

Table 1: Summary of D62 results. Masses in units of $m_p = 938.272\,\mathrm{MeV}$. PDG 2024 experimental values. Tensions use experimental uncertainties.

Quantity	PDL expression	PDL value	Exp. value	Status
M_Z/M_W	$1/\cos(19\pi/119)$	1.14049	1.13461	Theorem, 5182 ppm
v/m_p	$N_{\mathrm{tot}}^2/(R_e k_1)$	262.241	262.418	Conj., 676 ppm
M_Z/m_p	analytical chain	97.179	97.187	Conj., 3.67σ
M_W/m_p	$M_Z\cos(19\pi/119)$	85.208	85.656	Corollary
M_H/m_p	$\varphi k_2/\sin^2\theta_W + \Delta n\, k_2/N_{\mathrm{tot}}$	133.611	133.437	Conj., 1.49σ

This document makes five contributions to the PDL corpus.

Definition of gauge boson mass (Definition 2.1). The first formal definition of boson mass in the PDL programme, bridging the gap between the gauge group derivation (D57–D61) and the mass question. Gauge bosons are transport modes in $\mathbb{Z}_2^3$, not fermionic closures; their mass is a minimal excitation threshold, not a stationary coherence cost.

Derivation of the vev (Conjecture 3.3). The electroweak vev is expressed as a ratio of network capacities: $v = N_{\mathrm{tot}}^2/(R_e k_1)\cdot m_p$, with each factor justified by an unconditional theorem. The isolation of this candidate in the numerical search is robust.

The M_Z/M_W theorem (Theorem 5.1). The ratio $M_Z/M_W = 1/\cos\theta_W$ is an unconditional theorem of C1–C4. The tree-level PDL Weinberg angle agrees with the physical MS-bar value to 0.48σ.

The M_Z analytical chain (Conjecture 4.2). The chain $M_Z = g_Z(\Delta\alpha_{\mathrm{lep}} + \Delta\alpha_{\mathrm{had}})\cdot v/2$ yields 84.6 ppm at 3.67σ. The tension is precisely attributed: $\Delta\alpha_{\mathrm{had}} = \kappa_{\mathrm{surf}}/\varphi$ uses m_p as the lower integration bound, excluding pion and kaon contributions that will be recovered when OP-D59-1 is resolved.

The decay cascade (Observations 2.2, 7.1). The Higgs cascade is mapped exactly onto the $\mathbb{Z}_2^3$ topology. The four primary decay paths are derived combinatorially. Neutrinos exit the coherence network at level 2, carrying 12.2% of M_H and rendering the reconstructed LHC mass a coherent-fraction measurement.

The resolution of OP-D59-1 (quark and hadron mass spectrum from C1–C4) is the key next step: it will supply the missing pion and kaon contributions to $\Delta\alpha_{\mathrm{had}}$, derive

λ_H via the top quark Yukawa coupling, and enable D62's conjectures to be elevated to theorems.

References

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